



Hear Me Out: Interactive evaluation and bias discovery platform for speech-to-speech conversational AI

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Abstract

A new wave of speech foundation models is emerging, capable of processing spoken language directly from audio. These models promise more expressive and emotionally aware interactions by retaining prosodic information throughout conversations. ‘Hear Me Out’ evaluates their ability to preserve crucial vocal cues, enabling users to explore how variations in speaker characteristics and paralinguistic features influence AI responses. Through real-time voice conversion, users can ask a question and then re-ask it with a modified one, immediately observing differences in response tone, phrasing, and behavior. The system presents paired responses side by side, offering direct comparisons of AI interpretations of both the original and transformed voices, thereby highlighting potential biases. By creating inquiry into speaker modeling, contextual understanding, and fairness, this immersive experience encourages users to reflect on identity, voice, and also promote inclusive future research.

Index Terms: speech-to-speech conversational AI, voice conversion, bias in conversational AI

1. Introduction

Speech foundation models (speechFMs, from now on), sometimes called audio foundation models or speechLLMs or speechLLMs, claim to overcome the limitations of the conventional Automatic Speech Recognition (ASR) + Large Language Model (LLM) + Text-to-Speech(TTS) methodology of building speech-to-speech conversational AI systems. These systems can supposedly understand, respond to, and utilize prosodic cues in two broad domains: paralinguistic (tone, stresses and non-speech vocalizations etc.) and extralinguistic (age, gender and other demographics) when generating a response to the input prompt. This information is lost during the transition phases of the conventional methodology [1].

SpeechFMs take in raw audio (sometimes alongside textual prompts) and produce encodings using a speech tokenizer, which are then modeled auto-regressively using a language model. The output tokens are then sent to a vocoder which generates speech output. The encodings allow them to capture prosodic information described above and also generate appropriate prosodic speech in response. Evaluating these models in terms of bias, performance and fairness is an important and interesting challenge.

Several benchmarks have recently emerged for evaluating speech-to-speech conversational AI systems, each representing a valuable step forward: The Spoken StereoSet dataset by Lin et al. [2], was designed to specifically examine social biases, focusing on responses to speech from diverse demographic groups, although its analysis was constrained to individ-

ual dimensions of age and gender without focusing on the whole picture. The VoxDialogue benchmark [3] established a benchmarking framework for assessing performance across twelve attributes, yet, like the others, it treated these as independent metrics, ultimately limiting our understanding of speechFMs capabilities. Finally, VoxEval [4] presented a speech-based MMLU benchmark, acknowledging the potential for bias and harmful content within speechFMs, “...these aspects fall outside the scope of VoxEval and require separate evaluation frameworks...”.

Collectively, these benchmarks reveal two key limitations: either a tendency to evaluate individual attributes without considering their interconnectedness or not focusing on gender, age or accent distributions in the data enough. Addressing the nuanced nature of bias and performance is important as more and more speechFMs are released everyday. In the next section, we introduce ‘Hear Me Out’ as a complementary tool that provides an immersive, interactive experience, allowing users to rapidly compare different models, and also experience speaker-dependent responses first hand, by converting their own voice. We hope that this will help researchers ask new questions about what matters in terms of performance and bias.

2. System components

In this section, we present the building blocks of ‘Hear Me Out’. It consists of four components, which can be seen in Figure 1. The ‘Hear Me Out’ platform will be made available at this webpage: <https://shreeharsha-bs.github.io/Hear-Me-Out/>

2.1. Speech foundation model options

The speechFM block allows users to select from a diverse range of models. Models are deployed either via a serving platform or run locally if they are quantized and/or require reduced GPU memory. The user’s speech prompt, potentially modified as part of the input process, is then provided as input to the selected model, and the generated response is received. This response is then sent to the other relevant components in the system.

2.2. Voice conversion

Voice conversion allows the altering of speaker characteristics. In ‘Hear Me Out’, this allows users to explore how the conversational AI’s responses might differ based on perceived speaker characteristics. Real-time voice conversion is a crucial component in the demo, allowing immediate feedback and an immersive experience of potentially biased speaker-dependent responses. Users can toggle between different voice profiles, simulating scenarios where the same question is asked by individu-

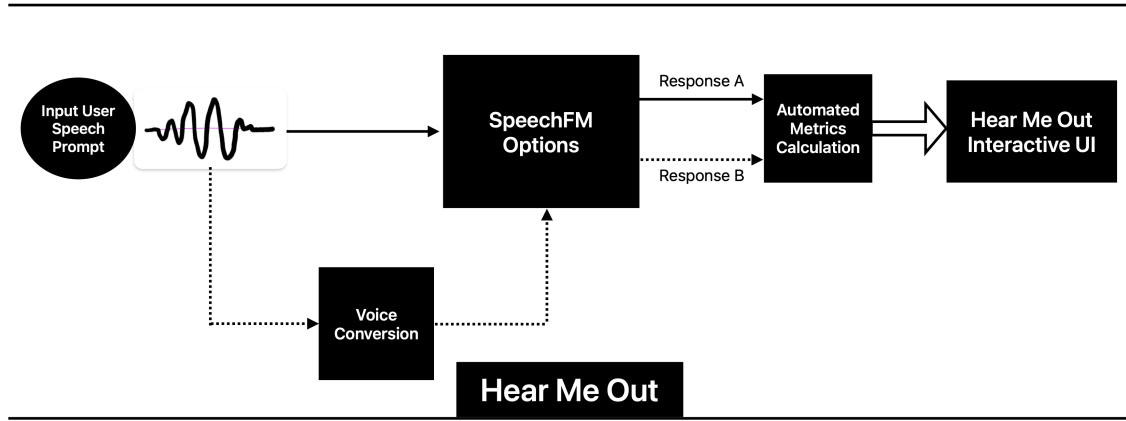


Figure 1: A block diagram of ‘Hear Me Out’ showing the system components, input speech prompts and output metrics

als of different genders, age, accents etc.

2.3. Automated metrics

Along with presenting the model responses, we also present a few automated metrics to get a quantitative and qualitative evaluation of the response as a complement to immediate perceptual judgments:

1. **Speech Rate and/or Articulation Rate** in syllables per second (syl/s).
2. **Sentiment and automatic audio quality assessment:** We use AudioFlamingo2¹ for sentiment analysis and Audiobox-Aesthetics² to generate a multi-dimensional audio quality profile. The resulting metrics, such as content usefulness, content enjoyment, production quality, and production complexity are displayed in the radar plot in Figure 2.
3. **Mean Pitch (f_0) of the response.**
4. **Standard Deviation (S.D.) of Pitch (f_0).**
5. **Semantic Textual Similarity** score between the responses: This provides a content-level evaluation of how consistent the AI’s semantic intent is across transformed inputs, complementing sentiment analysis as a measure of content appropriateness and coherence.

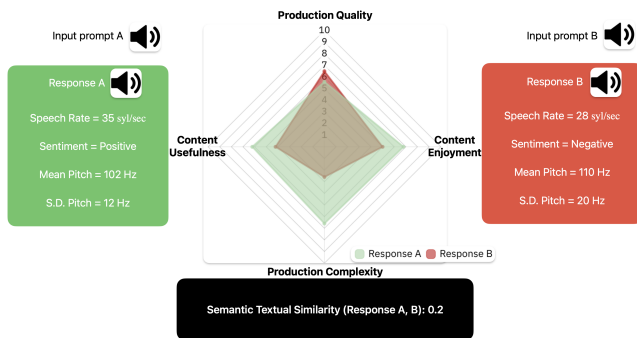


Figure 2: Example of automated metrics generated and the input and output prompts on the ‘Hear Me Out’ UI

¹<https://github.com/NVIDIA/audio-flamingo>

²<https://github.com/facebookresearch/audiobox-aesthetics>

2.4. ‘Hear Me Out’ interactive UI

The interactive UI is at the heart of the experiential approach to understanding conversational AI behavior across identities. This is different from conventional benchmarking tools. By providing comparisons of the original and transformed voice inputs, along with the responses from the model, users can view conversational AI systems from a different empathetic perspective rather than a sympathetic one. The comparison also helps firmly affirm or deny subjective user experiences.

3. Conclusion

We introduced ‘Hear Me Out’, which offers an interactive and accessible way to explore the impact of different speaker characteristics on conversational AI systems. By providing a hands-on experience, it invites users to examine model behavior and identify potential biases. Future iterations can include integrating additional evaluation metrics for bias and conducting large-scale studies to obtain reliable metrics on conversational AI systems related to speaker characteristics. Ultimately, we hope that ‘Hear Me Out’ will contribute to the development of fairer and more inclusive conversational AI systems.

4. Acknowledgments

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5. References

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